

Speech and Audio Systems

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of speech pipelines and multimodal evidence.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Examples and Rubrics. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 12: Speech and Audio Systems

Speech interaction includes capture, recognition, interpretation, and response stages.



CREATOR MISSION

Trace errors from noisy audio to wrong action.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Image generation and provenance. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week's end goal

By the end, I can separate speech recognition, calculation and speech output when diagnosing an error.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use speech and audio systems to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

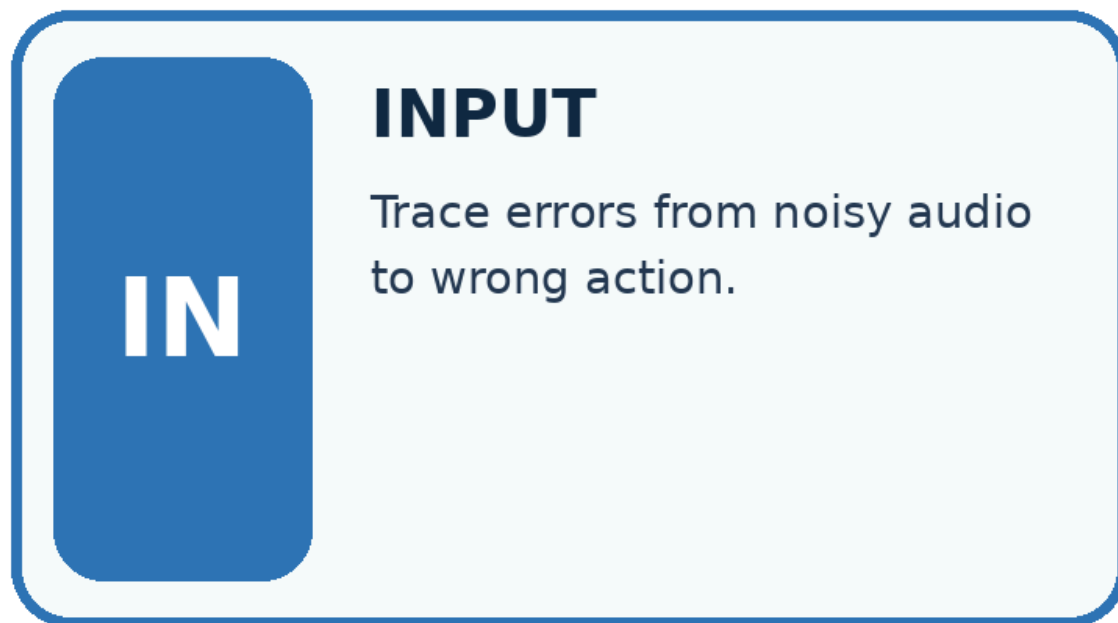
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Speech interaction includes capture, recognition, interpretation, and response stages.



OUT

EVIDENCE

Trace errors from noisy audio to wrong action.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Track source modality, segment/time references, transcription, interpretation and final claims separately. Compare conflicting evidence without assuming one modality always wins. Ask for clarification when the task needs unresolved information.

Speech pipelines and multimodal evidence — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Check or calculation: Track source modality, segment/time references, transcription, interpretation and final claims separately. Compare conflicting evidence without assuming one modality always wins. Ask for clarification when the task needs unresolved information.

Conclusion and limit: There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Correct the transcript to “thirteen plus seven” with all later stages unchanged.

Prediction: The calculator and spoken result should become 20 under the declared pipeline.

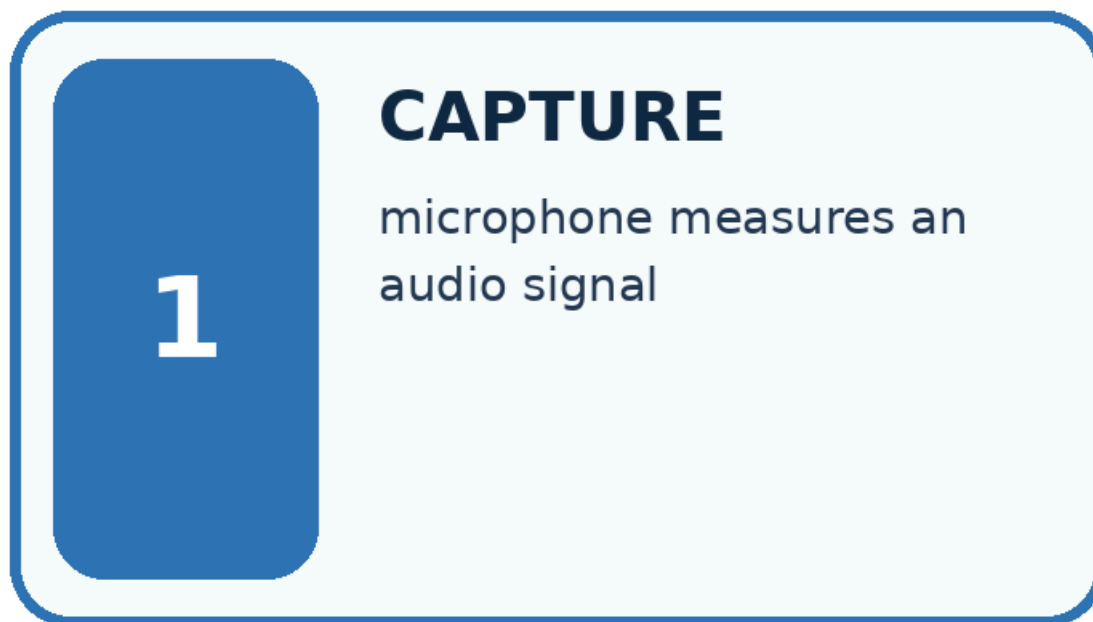
Reason: $13+7=20$; the changed input removes the earlier recognition error.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

SPEECH RECOGNITION

signal becomes candidate words

3

INTENT

system's interpretation of the request

4

SYNTHESIS

text becomes generated
speech

Word · My own example · Not this / boundary

Word: capture

My own example: Record a signal, observation or item of information for later use.

Not this / boundary: _____

Word: speech recognition

My own example: Converting spoken audio into an estimated sequence of words.

Not this / boundary: _____

Word: intent

My own example: The goal a user is trying to achieve, which may need clarification.

Not this / boundary: _____

Word: synthesis

My own example: Combining information into a new account or output while preserving its meaning and support.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Modality

Meaning: A form of information such as audio, image or text

Word: Transcription

Meaning: A text representation of speech

Word: Error propagation

Meaning: An earlier error affecting later processing

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 12 • WORKED EVIDENCE

- 1 Learner says 'define erosion' in a noisy room.
- 2 Recognizer outputs 'design erosion.'
- 3 Intent stage follows the wrong transcript.
- 4 Display transcript, allow correction, and avoid high-consequence action from uncertain audio.

1. Learner says 'define erosion' in a noisy room.

2. Recognizer outputs 'design erosion.'

3. Intent stage follows the wrong transcript.

4. Display transcript, allow correction, and avoid high-consequence action from uncertain audio.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Speech and Audio Systems

A fictional learner dictates a calculation to a voice assistant.

The facts we will use

Known audio phrase: “thirteen plus seven”. Transcript: “thirty plus seven”. Calculator returns 37.

Synthetic voice says “thirty-seven”. The example supplies ground truth; no real audio was analysed.

Speech and Audio Systems

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Recognise speech

Transcription changes thirteen into thirty.



2. Calculate text

$30+7=37$ is correct for the mistaken transcript.



3. Speak result

Speech synthesis repeats 37; fluent voice does not validate the input.



4. Confirm intended input

Read back the recognised expression; intended $13+7=20$.

CHANGE ONE THING

Correct the transcript to "thirteen plus seven" with all later stages unchanged.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Recognise speech

What happens and why: Transcription changes thirteen into thirty.

Step: 2. Calculate text

What happens and why: $30+7=37$ is correct for the mistaken transcript.

Step: 3. Speak result

What happens and why: Speech synthesis repeats 37; fluent voice does not validate the input.

Step: 4. Confirm intended input

What happens and why: Read back the recognised expression; intended $13+7=20$.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The first error occurs before arithmetic. Blaming the calculator would target a stage that correctly processed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action. Retain the original audio and transcript, where permitted, to separate recognition defects from later processing errors.

What this example does not establish

These supplied transcript facts do not establish how an actual recogniser performs for accents, noise or different speakers.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A wrong spoken response means the language model caused the first error. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project,

select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A stage-level accuracy or authority policy does not establish every modality agrees or every downstream claim is supported.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Speech pipelines and multimodal evidence”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Speech pipelines and multimodal evidence. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which stage was initially wrong, and which stage was correctly operating on a wrong input?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Trace four audio failures stage by stage, locate the first mismatch, and propose a stage-specific test or fallback.

Trace four audio failures stage by stage, locate the first mismatch, and propose a stage-specific test or fallback.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction:

Actual:

Evidence / calculation:

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what

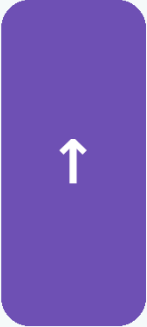
did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY
1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Show transcripts and require confirmation before saving memory or using tools.

THIS WEEK’S CHECKPOINT Show transcripts and require confirmation before saving memory or using tools.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

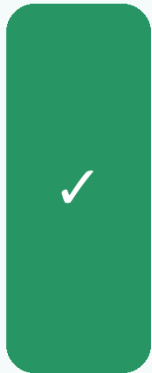
Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Trace errors from noisy audio to wrong action.

CHALLENGE Trace errors from noisy audio to wrong action.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 12 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A speech tool hears fifteen as fifty, then calculates correctly using fifty.

The question

Where did the error enter?

Worked explanation

Transcription supplied the wrong number. Confirm the transcript or ask for clarification. Correct arithmetic cannot repair wrong input.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Trace a speech error through the system

Evidence or stage · Value or result

Evidence or stage: Speech

Value or result: Fifteen

Evidence or stage: Transcript

Value or result: Fifty, the first error

Evidence or stage: Calculator

Value or result: Correct arithmetic using fifty

Evidence or stage: Result

Value or result: Wrong for the original request

A speech tool hears fifteen as fifty, then calculates correctly using fifty.

Where did the error enter?

Transcription supplied the wrong number. Confirm the transcript or ask for clarification. Correct arithmetic cannot repair wrong input.

14. Try a new case independently

Independent case: Changed independent case

Known audio says “nineteen minus six”. Transcription says “ninety minus six”; the calculator returns 84. Locate the first error and propose a check.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 12 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A recording says fifteen, but the transcript says fifty. The summary repeats fifty. Where did the error enter, and what should be checked first?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Show transcripts and require confirmation before saving memory or using tools.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Audio says 'sixteen divided by four'; transcript says 'sixty divided by four'; calculator returns 15. Which stage first fails? What would a useful confirmation show?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a

model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.